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to compensate for the unreliable timing of the network, these systems require both controlled latencies and guaranteed bandwidth.

IEEE 802.1 Audio Video Bridging (AVB) Task Group (renamed Time-Sensitive Networking Task Group) is working on standards to meet requirements such as 802.1Qat (Stream Reservation), a simple reservation protocol to notify various network elements in a path to reserve the resources necessary to support a particular stream.

801.1Qav (queuing and forwarding for AV bridges) defines the rules to ensure that an AV stream will pass through the network within the delay specified in the reservation.

IEEE also standardised IEEE 1722 (layer 2 transport protocol for time-sensitive applications in bridged local area networks) and 1733 (layer 3 transport protocol for time-sensitive applications in local area networks). These protocols use time (from IEEE802.1AS) as well as worst-case transport delay to get a presentation time, which is inserted in the packet.

1733 is used with RTP/RTCP and 1722 is used with IEC 61883 and other formats. AVnu Alliance is an industry forum dedicated to the advancement of AV transport through the adoption of IEEE 802.1 AVB and related IEEE 1722 and 1733 standards. This alliance is heavily supported by the automotive industry.

Test requirements

PHY testing. Testing has to be conducted in both 100Mbps (BroadReach) and 1GE (802.3bp 1000BASE-T1) PHYs. Physical testing might have a variety of purposes, so that requirements of a specification are met. Testing demonstrates proof of concept, demonstrates utility of a proposed implementation, provides standard data for engineering or quality-assurance purposes and validates suitability for end use.

Switch testing. Automotive Ethernet switches will need to test the new AVB and IET support capabilities. The test such as queue-reservation protocols, getting expected packet loss, bandwidth and latency for each class of service, behaviour test when a device connected to the switch is misbehaving, test for failover and convergence times, negative test cases and 2554/2889 testing with the above conditions are taken.

Security testing. Cars are becoming more connected and software-driven. Security is becoming more critical than ever before. Car manufacturers need to ensure that cars have multiple fail-safe systems to prevent intrusion, especially when it comes to power train domain.

Challenges ahead

One of the biggest challenges in creating automotive-grade Ethernet was finding an alternative to the multi-pair unshielded twisted-pair cabling commonly used by 100BASE-T and 1000BASE-T networks. Category-5/6 cabling is simply too bulky and expensive to be practical in the tight confines and tight budgets of high-production automobiles.

In order to meet the unique cost and manufacturability requirements of automotive applications, OPEN (One-Pair Ether-Net) Alliance Special Interest Group has developed a new PHY technology. It is being called 100BASE-T1 or 1 Twisted Pair 100Mbps Ethernet (1TPCE). It uses digital echo cancellation and decision feedback equalisation (DFE) techniques borrowed from 1000BASE-T Gigabit Ethernet to support full-duplex Ethernet at 100Mbps over a single pair of unshielded twisted wires.

Meanwhile, more widespread use of automotive Ethernet could, in theory, increase the potential for hacking through a maintenance interface or in-car Wi-Fi hotspots. **EFY**